

CLASSIFICATION

CLASSIFICATION TECHNIQUES – PART V

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CLASSIFICATION TECHNIQUES

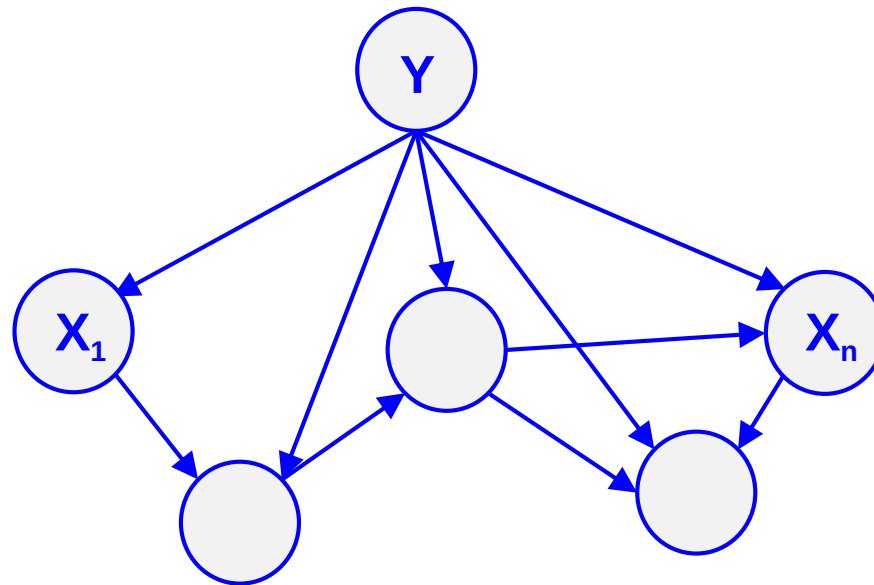
The following concepts will be introduced:

- ✓ **BAYESIAN NET/BAYESIAN NET CLASSIFIERS**
- ✓ **TREE AUGMENTED NAÏVE BAYES**

PROBABILISTIC: BAYESIAN NETS AND OTHERS 1

The **NAIVE BAYES CLASSIFIER** is generalized by **BAYESIAN NETWORKS** which make less drastic conditional independence assumptions but exploit the concept of **SPARSITY**.

An example of **BAYESIAN NETWORKS CLASSIFIER** is the following.



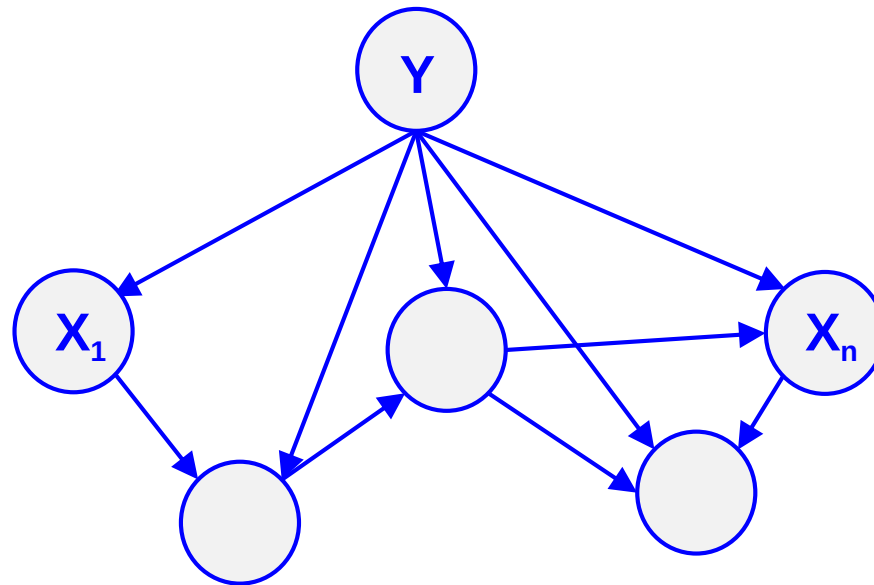
The **CLASS ATTRIBUTE** is still associated with the **ROOT NODE Y**.

The **EXPLANATORY ATTRIBUTES** are associated with nodes having not only the **ROOT NODE Y** as their incoming arcs, **EXPLANATORY ATTRIBUTES** can point to other **EXPLANATORY ATTRIBUTES**.

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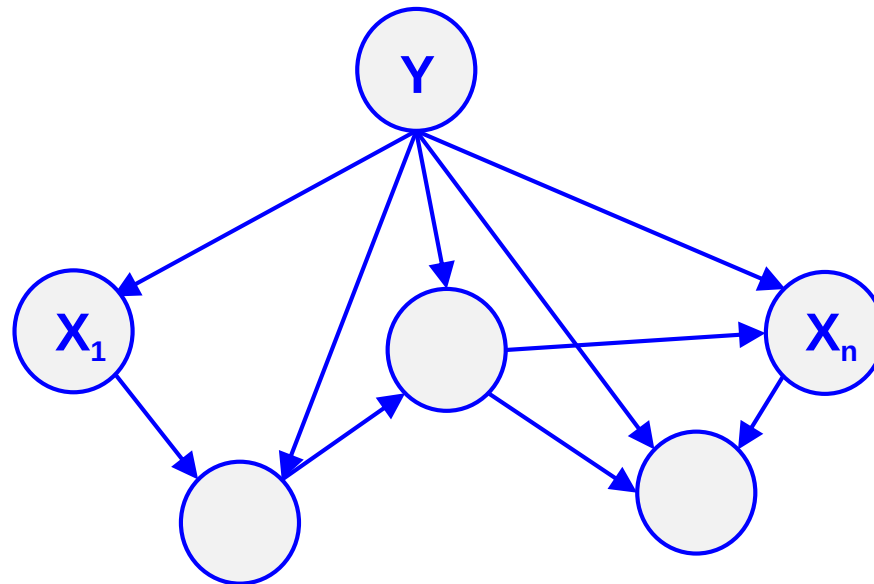


NO CYCLES ARE ALLOWED, a constraint of **BAYESIAN NETS**.

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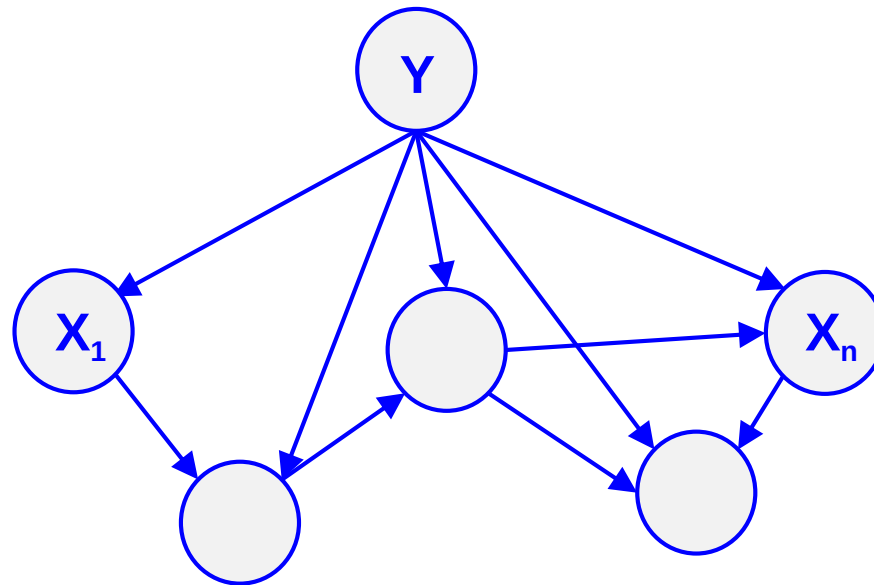
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This generalization **KEEPS INTO PROPER ACCOUNT DEPENDENCIES BETWEEN** the **EXPLANATORY ATTRIBUTES** that are **NO LONGER ASSUMED TO BE INDEPENDENT GIVEN** the **CLASS ATTRIBUTE Y**.

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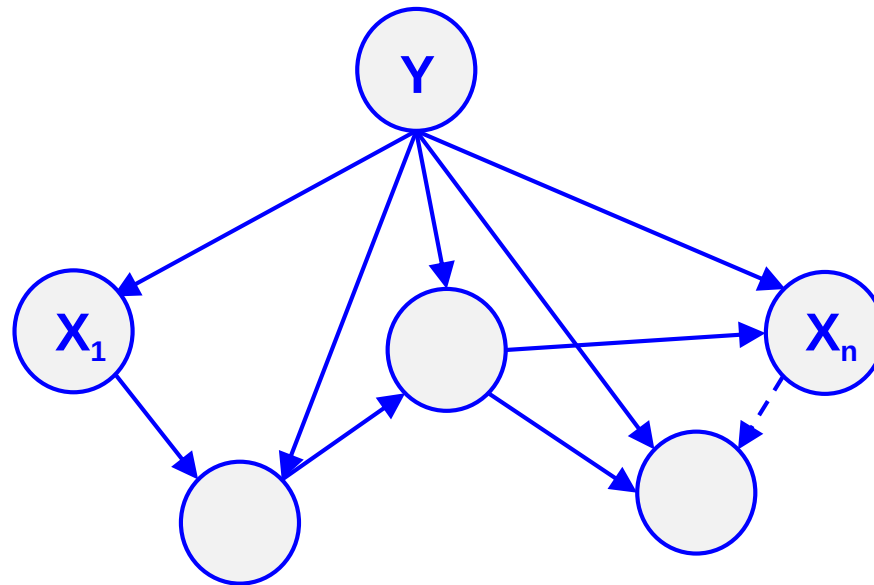
An example of **BAYESIAN NETWORKS CLASSIFIER** is the following.



Many modifications of **BAYESIAN NET CLASSIFIERS** have been proposed in the specialized literature.

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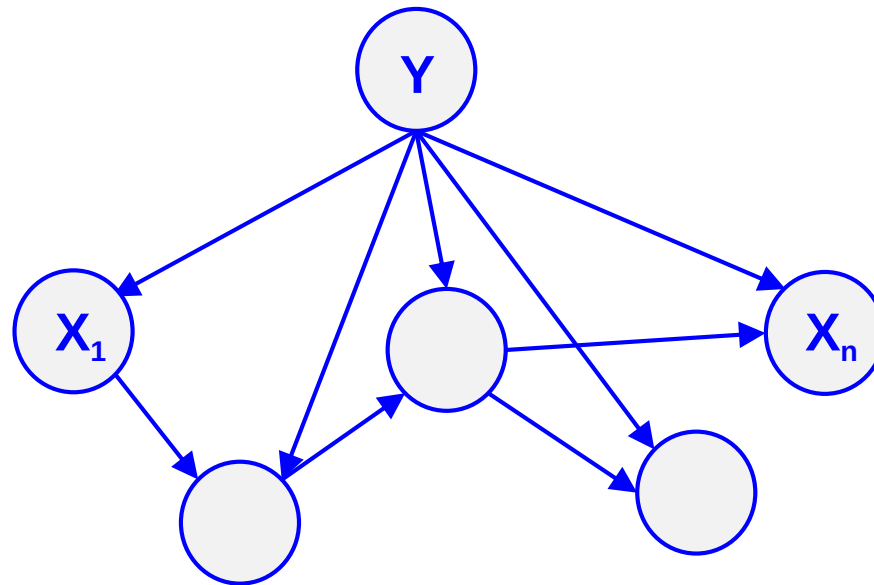


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Removing the dashed arc from the Bayesian Net in figure brings to **TREE-AUGMENTED NAIVE BAYES**.

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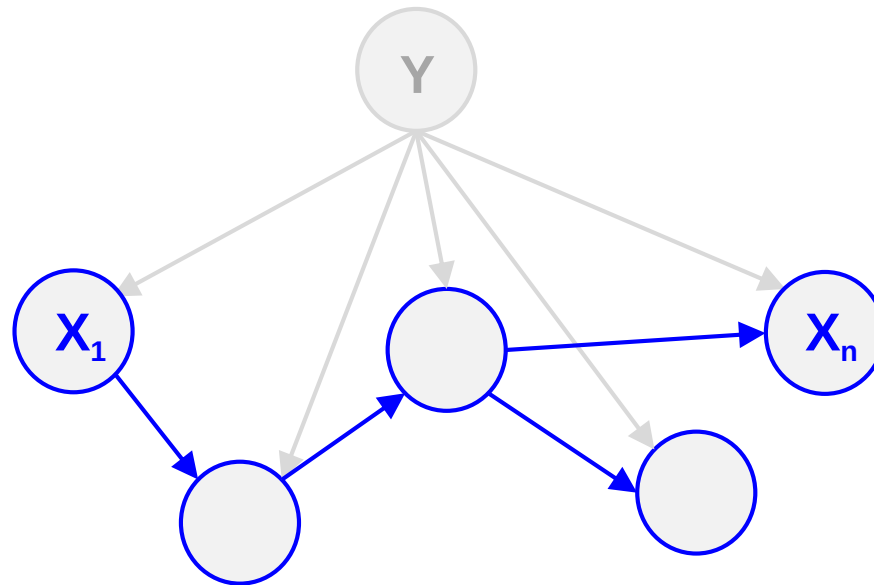
An example of **BAYESIAN NETWORKS CLASSIFIER** is the following.



Each node has at most one Explanatory Attribute as its parent (incoming arc from), thus if we remove the root node a tree is obtained which gives the name to this particular type of Bayesian Net Classifier.

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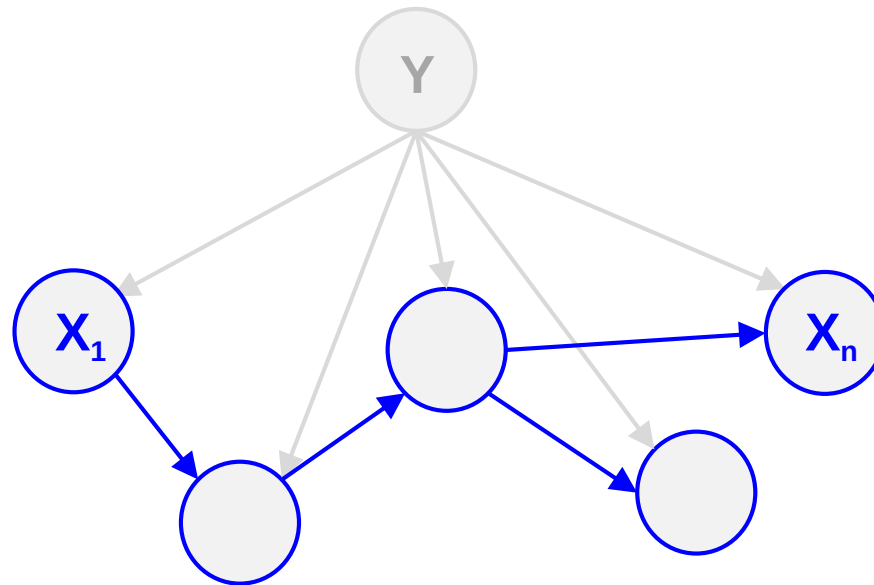
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An example of **BAYESIAN NETWORKS CLASSIFIER** is the following.



Many other modifications include **AODE**, **HNB**, ... All oriented toward the implementation of a trade-off between expressiveness and complexity.

They can be applied to different types and combinations of Explanatory Attributes and type of Class Attribute.

	Explanatory Attribute				Class Attribute	
	Categorical			Numeric	Categorical	
	<i>Nominal</i>	<i>Ordinal</i>	<i>Binary</i>		<i>Nominal</i>	<i>Binary</i>
<i>Decision Tree</i>	X	X	X	X		X
<i>Logistic Regression</i>				X		X
<i>SVM</i>		X	X	X		X
<i>MLP</i>		X	X	X		X
<i>NB</i>	X	X	X	X	X	X

However, is it possible to transform nominal Explanatory Attributes and nominal Class Attributes to apply the classification techniques that can not allocate nominal Attributes.

Please note that **CLASSIFICATION TECHNIQUES ARE NOT THE BEST OPTION WHEN AN ORDINAL ATTRIBUTE IS TO BE PREDICTED.**